Unsupervised Transformation

Instructor: Chao Huang

Department of Computer Science
Montclair State University

Main sources: CS@CMU&UT

Types of Unsupervised Learning

- **Clustering**
  - Group data into sets of items similar to each other
  - Example
    - ★ Group customers based on purchasing patterns
- **Unsupervised transformation**
  - Create new data representations easier to understand
  - Example
    - ★ Dimensionality reduction of a gene dataset to a few key components

Dimensionality Reduction

- High dimensions $\rightarrow$ many features

Document classification

Features per document =
thousands of words/unigrams
millions of bigrams, contextual information

Surveys - Netflix

480189 users x 17770 movies

	movie 1	movie 2	movie 3	movie 4	movie 5	movie 6
Tom	5	?	?	1	3	?
George	?	?	3	1	2	5
Susan	4	3	1	?	5	1

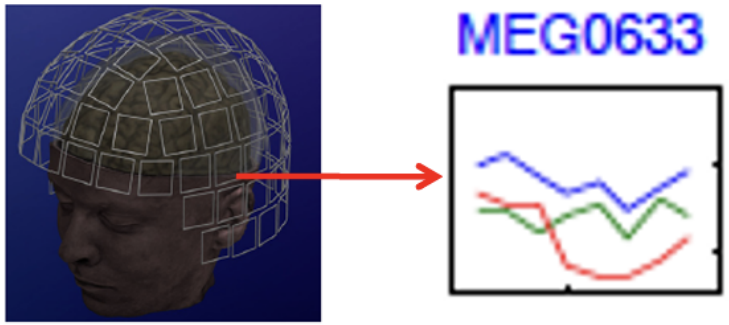
Dimensionality Reduction - Cont'd

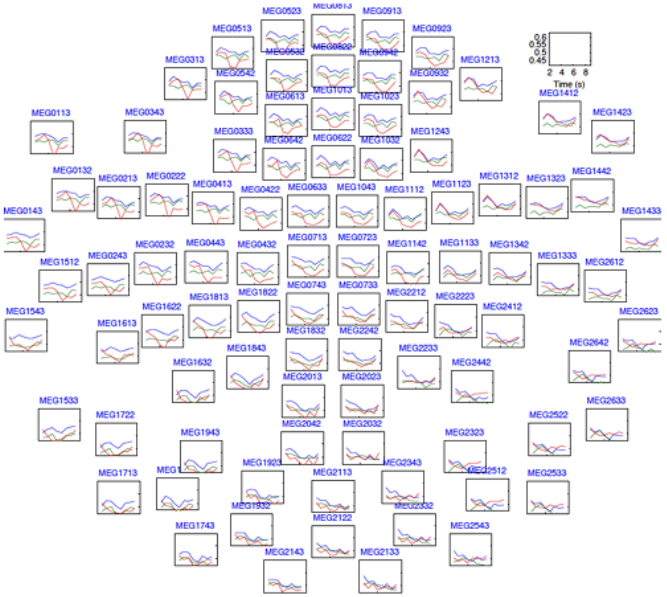
- High dimensions $\rightarrow$ many features

MEG Brain Imaging

120 locations x 500 time points

x 20 objects





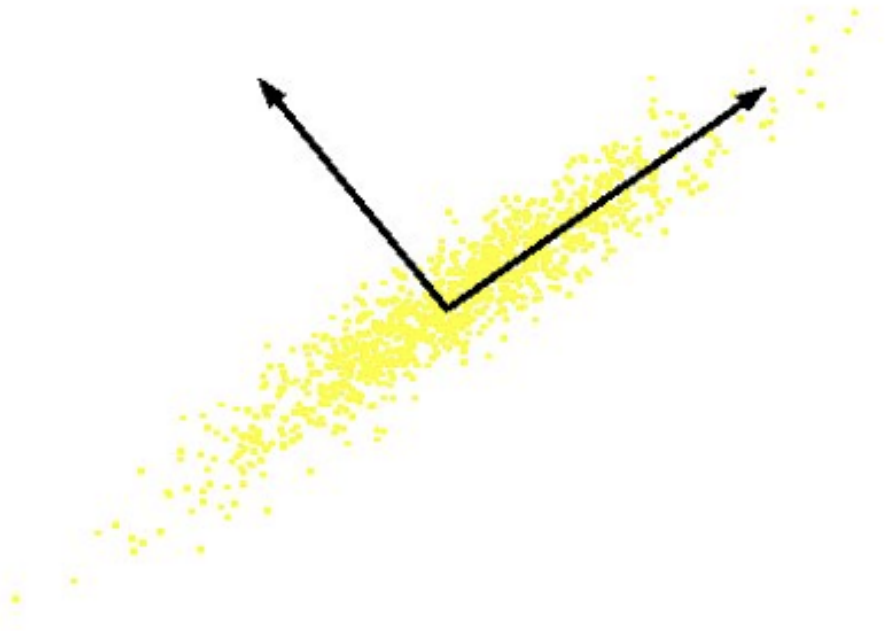
Or any high-dimensional image data

Dimensionality Reduction - Why?

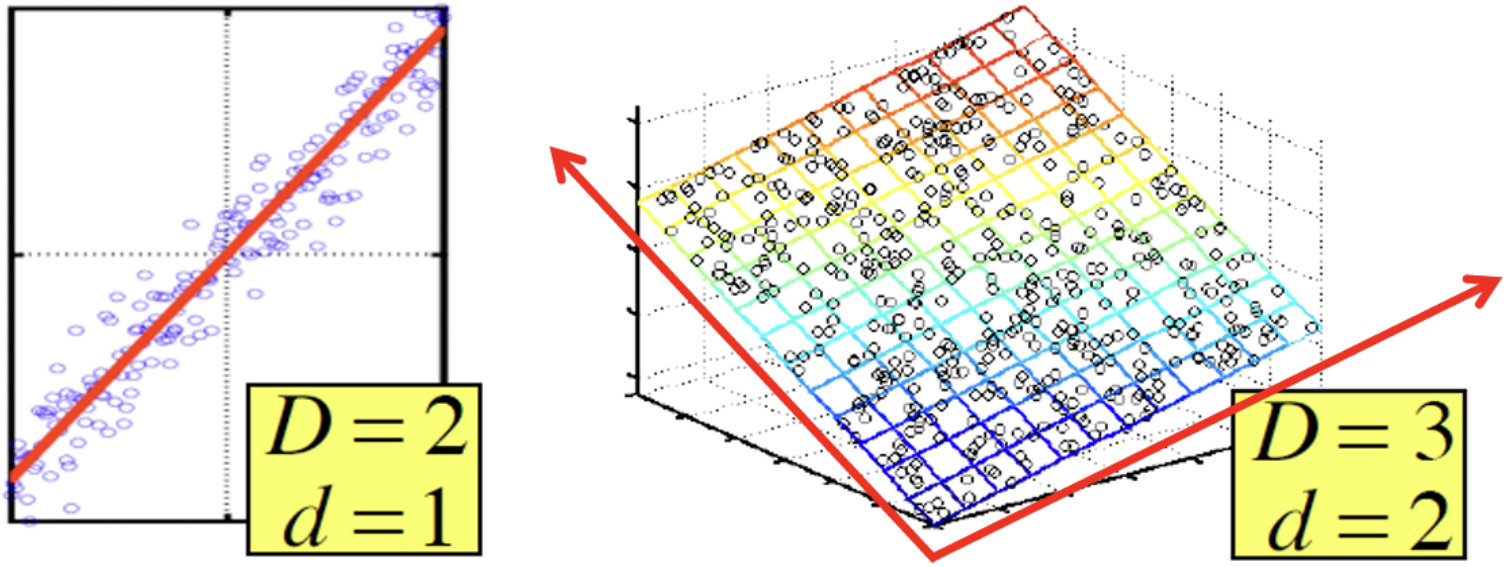
- Visualization
- More efficient use of resources (e.g., time, memory)
- Noise removal
- Further processing by ML algorithms

Principal Component Analysis (PCA)

- PCA is an unsupervised technique for extracting variance structure from high dimensional datasets
  - Intuition: variance contains more information
- An orthogonal projection or transformation of the data into a (possibly lower dimensional) subspace so that the variance of the projected data is maximized



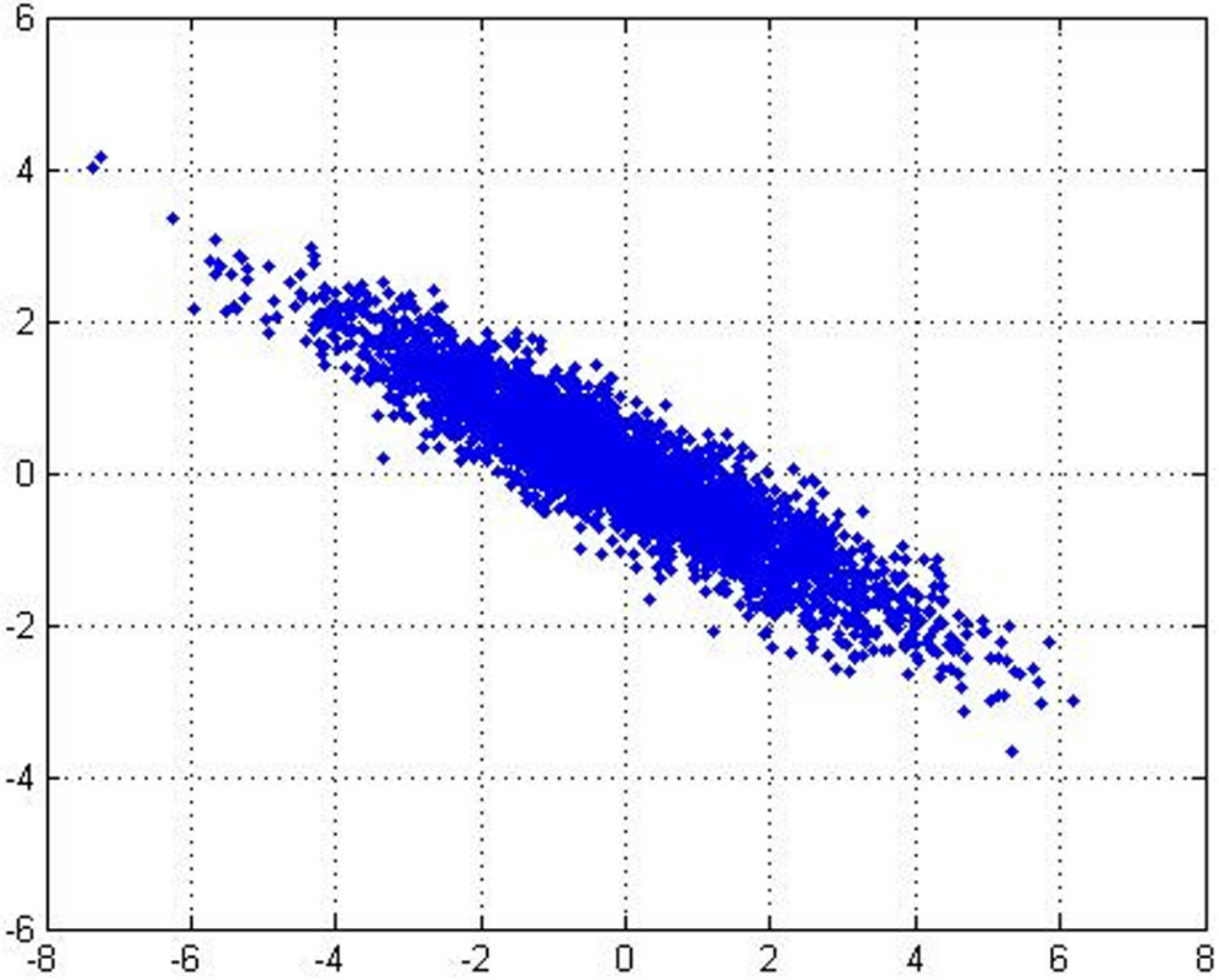
PCA - Cont'd



- If data lies on or near a low d-dimensional linear subspace, axes of this subspace are an effective representation
- Identifying the axes is known as Principal Component Analysis
  - Can be obtained by using classic matrix computation tools, e.g., eigen/singular value decomposition

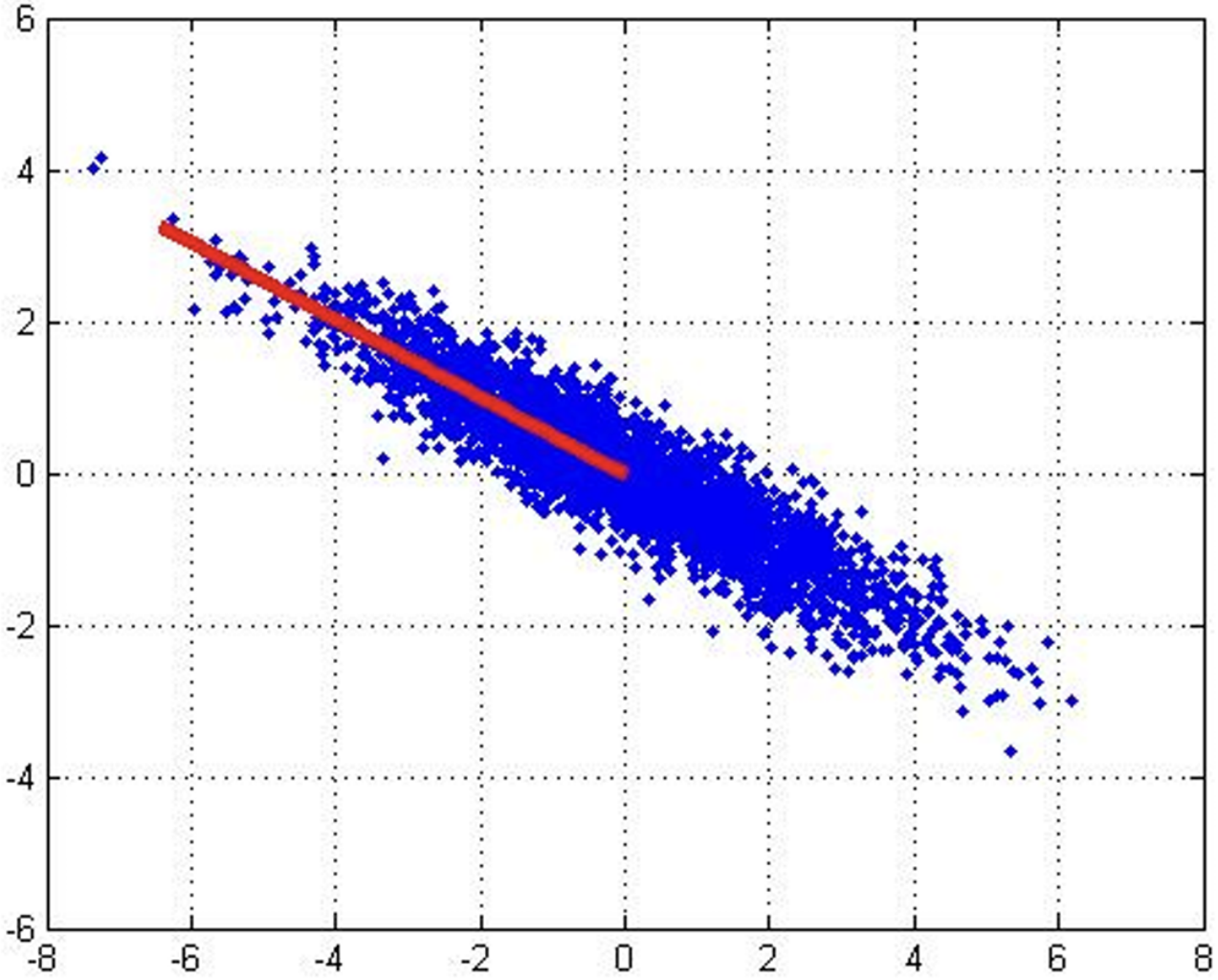
Example

- Given a 2d dataset, find the first and second PCA axes



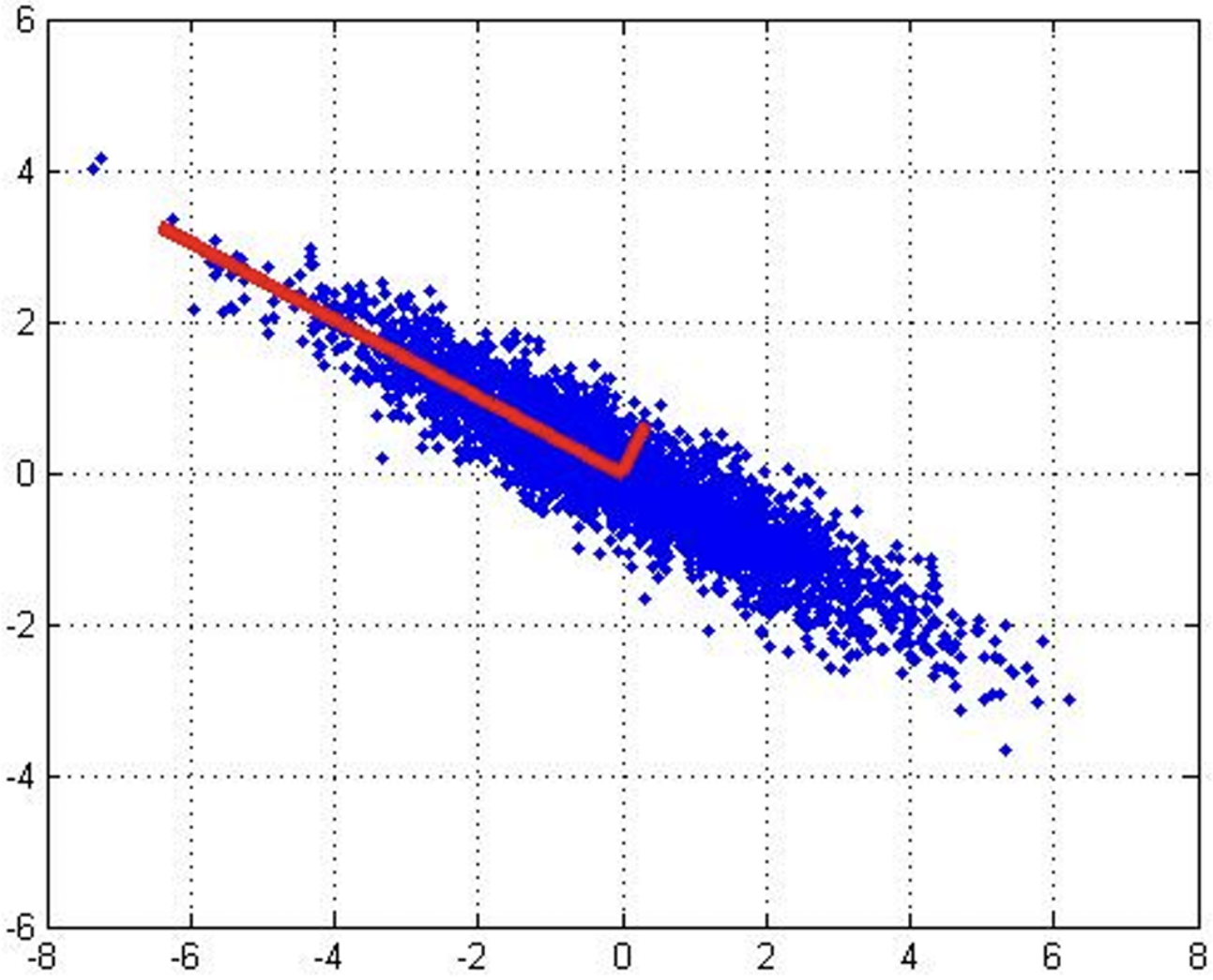
1st Axis

- Given a 2d dataset, find the first and second PCA axes



2nd Axis

- Given a 2d dataset, find the first and second PCA axes



t-SNE

- t-SNE: t-distributed stochastic neighbor embedding
- Idea: find a low-dimensional representation of the data that preserves the distances between points as best as possible

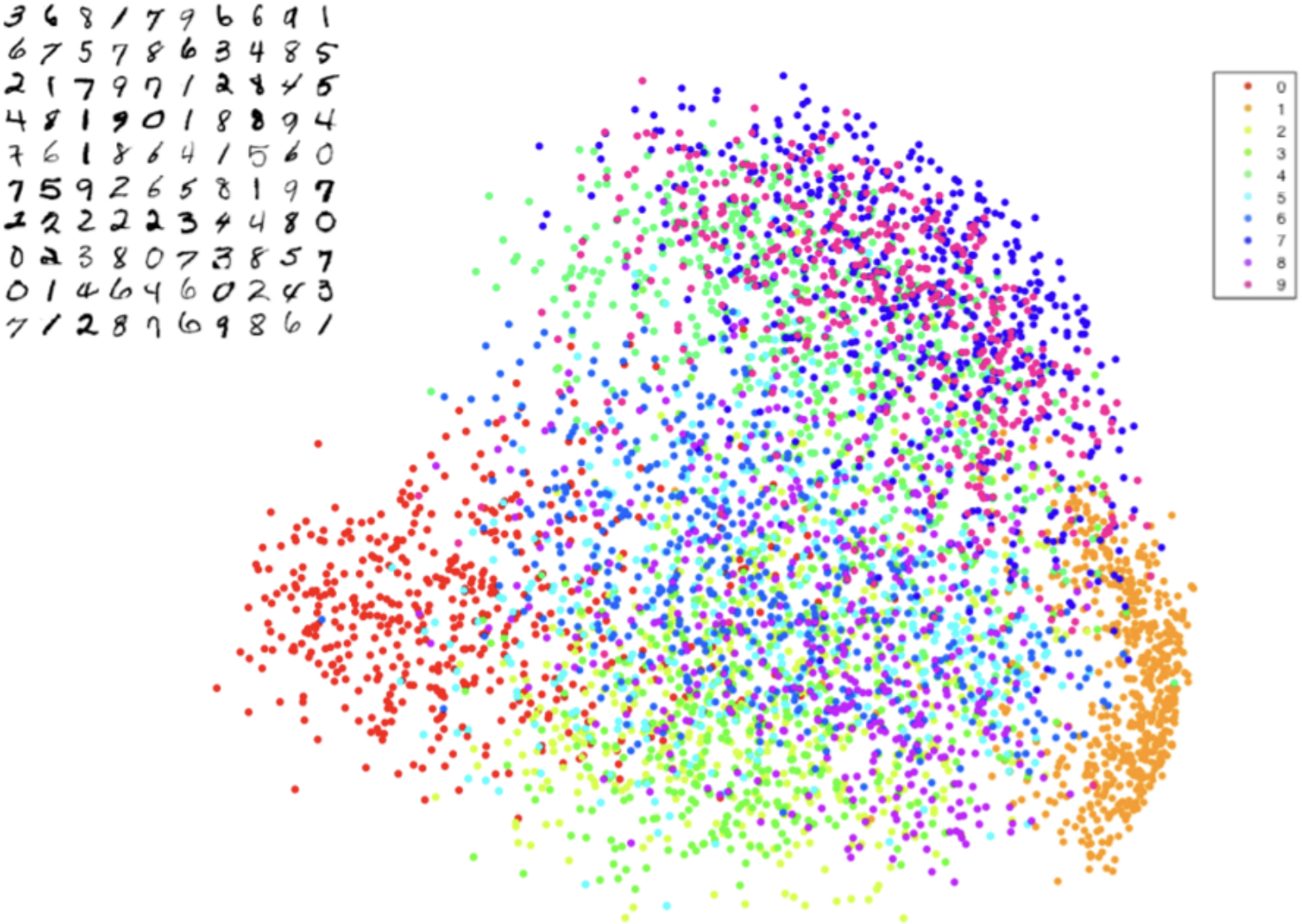
MNIST

- Dataset: MNIST
- Question: what is the dimension of MNIST?
  - For tabular data: dimension $\rightarrow$ # features
  - For image data: dimension $\rightarrow$ # pixels

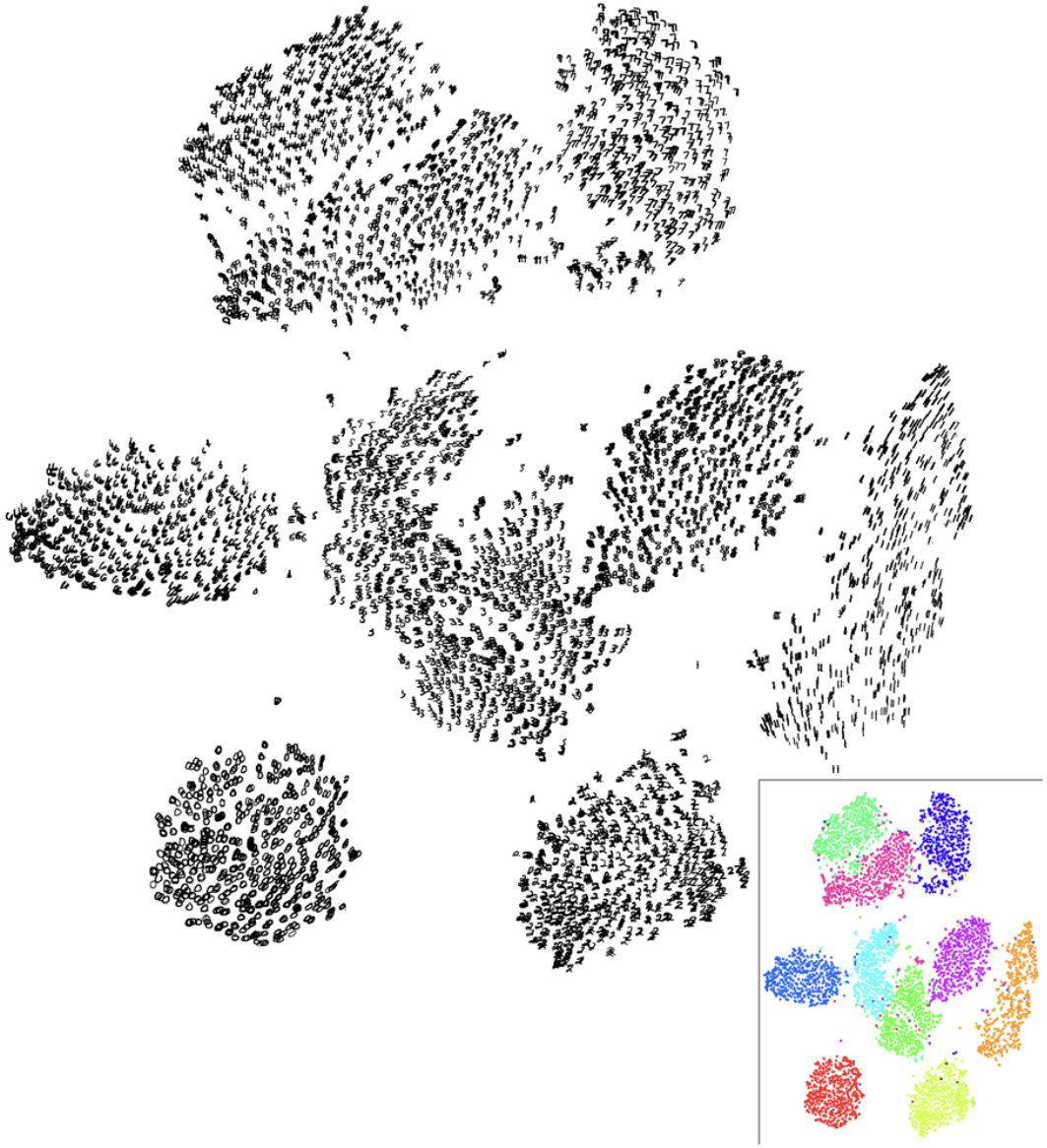
MNIST

- Dataset: MNIST
- Question: what is the dimension of MNIST?
  - For tabular data: dimension $\rightarrow$ # features
  - For image data: dimension $\rightarrow$ # pixels

2D Embeddings: PCA



2D Embeddings: t-SNE



Summary

- Dimensionality reduction
  - PCA: reduces dimensionality by maximizing variance along principal components and is useful for preserving global structures in data
  - t-SNE: focuses on preserving local relationships, ideal for visualizing complex high-dimensional data in a low-dimensional space, such as 2D or 3D. It tends to be more computationally expensive but better at capturing complex structures

After-Class Reading

- Müller A C, Guido S. Introduction to machine learning with Python: a guide for data scientists[M]. " O'Reilly Media, Inc.", 2016.
  - Sec. 3.4